

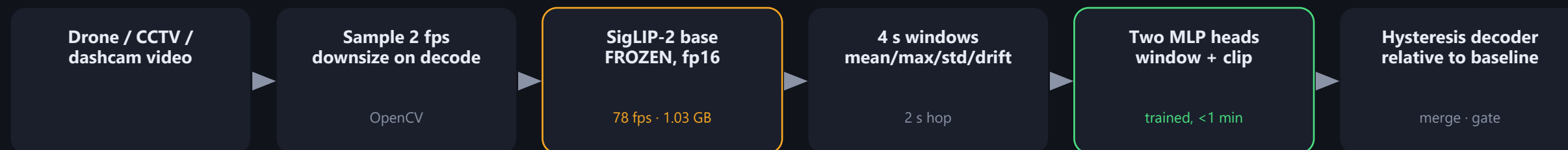
Video anomaly detection on a 4 GB laptop GPU

Frozen vision-language encoder + two small trained heads · 11 classes · 20.3x real time on an RTX 3050 Laptop

59.5 / 100

PROJECTED MARKS · PUBLIC TEST

PIPELINE



Encode once, reuse forever.

The encoder is frozen, so all 3,207 videos are embedded once into a cache. Retraining a head then takes under a minute, which is what made same-day iteration on windowing and thresholds possible.

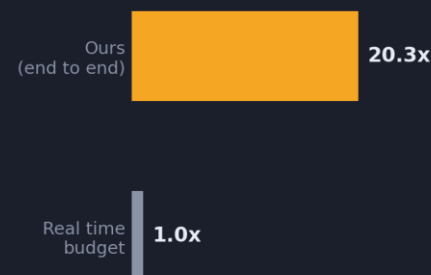
WHY NOT RUN A VLM PER FRAME

4 GB rules out fine-tuning a 2B VLM locally. So we froze one and trained only what sits on top.

The always-on stage is, in the end, a 12-way decision — orders of magnitude cheaper as a head over frozen embeddings than as generative decoding. The open-vocabulary semantics still come from the VLM; only the task-specific part is learned.

Explanations use nearest-neighbour retrieval over training descriptions in the same embedding space — one dot product per event, no generative model.

SPEED, MEASURED END TO END



3,391 s
of video

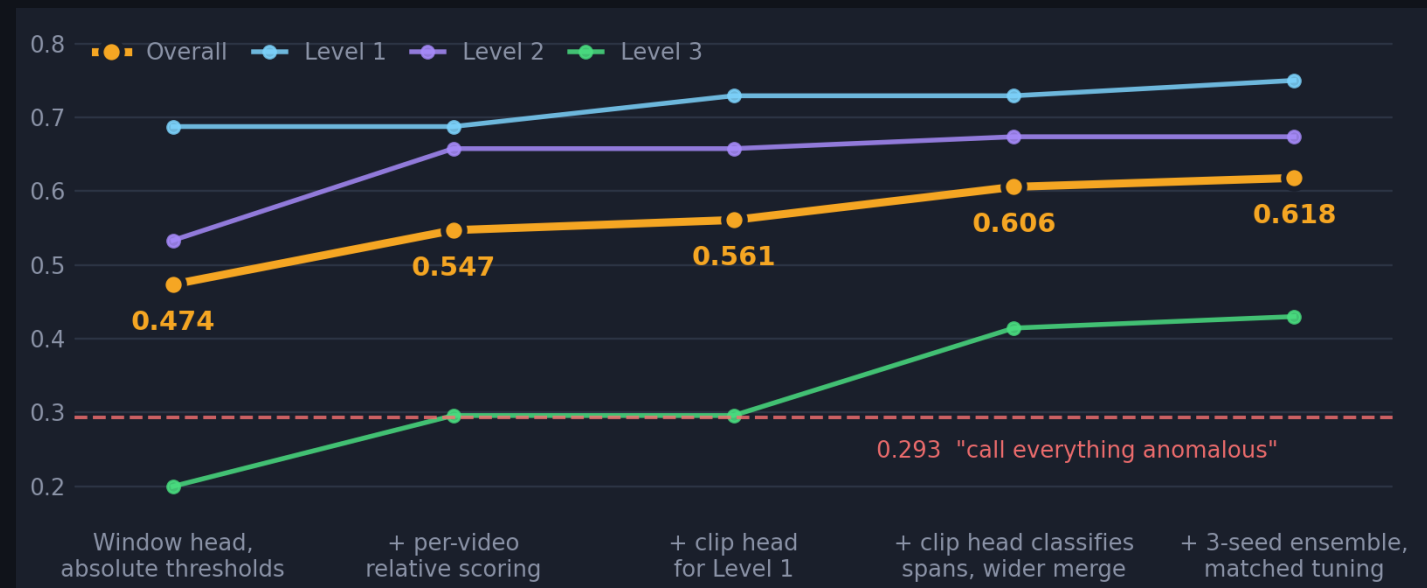
167 s
to process, incl. decode

Decode is the bottleneck, not the GPU.

Encoding 2 fps costs 1.03 GB and leaves headroom for many streams.

What moved the score, and what didn't

Every number below is the same 34-video public test set, scored locally against a re-implementation of the arena metric



FINAL, BY DIFFICULTY TIER

Difficulty 1

0.750

18.8 / 25 marks

anomaly acc 23 of 24

Difficulty 2

0.674

23.6 / 35 marks

both normals silent

Difficulty 3

0.430

17.2 / 40 marks

weakest, and worth most

Honest caveat.

The decoder is tuned on 6 Difficulty-2 and 4 Difficulty-3 videos, so those thresholds are the least trustworthy part of the system, and ties break toward the least aggressive setting. Our scorer matches the published structure of the metric; the exact within-tier weighting isn't published, so it ranks configurations rather than predicting the leaderboard.

THREE THINGS THAT MATTERED

Score against each video's own baseline

The head, trained on short clips where the event fills the frame, reports a high flat pedestal on long footage — baselines ranged 0.55 to 0.88, so no absolute threshold transfers. Thresholding on the rise above each video's own median found the events.

L2 +0.12 · L3 +0.10

Use each head for its own question

The window head answers where; the clip head answers what. Letting the clip head classify each detected span, instead of pooling window votes, fixed whole videos at once.

L3 0.296 → 0.414

Merge gaps sized to real events

Level-3 events run 45-125 s. A 5 s merge ceiling shattered one event into 13 fragments — and only the best-overlapping fragment can ever match.

T031 0.200 → 0.848

WHAT WE TRIED AND DROPPED

Larger SigLIP (segfaults at 4 GB) · zero-shot prompt fusion (0.29 alone, no lift) · one shared threshold for all tiers